

Olusegun Odeyemi

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Professional Profile

Security Engineer & Technical Founder specialising in advanced risk intelligence, automated compliance architecture, and secure Web3 integrations.

A hands-on technical architect bridging machine learning, decentralised systems, and enterprise cyber-security. I design and build secure-by-default infrastructure that scales to process complex transaction flows and dynamically assesses systemic risk. My expertise spans building full-stack compliance engines (Next.js, FastAPI, Flutter), writing smart contracts for privacy-preserving verification (ZK-SNARKs), and implementing data-driven anomaly detection models capable of massive transaction classification. Recognised for pioneering the Blind-Spot Decomposition Theory (BSDT) to mathematically map system failures and building the Anti-Money Laundering Transaction Transfer Protocol (AMTTP).

Core Competencies

Security Engineering & Technical Architecture

- Secure-by-design application development, API gateway security, and zero-knowledge identity protocols
- Hardening cloud infrastructure and containerised deployments across distributed microservices
- Penetration testing (black-box), vulnerability exploitation, and programmatic remediation of OWASP Top 10 flaws

Machine Learning & Automated Fraud Detection

- Engineering high-accuracy predictive models (XGBoost, Graph ML) for real-time anomaly detection
- Processing and classifying millions of concurrent data payloads with high inference speeds
- Architecting automated regulatory policy engines that adapt to high-velocity transactional environments

Data Systems & Secure Enterprise Integrations

- Managing secure multi-million revenue data flows and third-party API dependencies
- Implementing robust Role-Based Access Control (RBAC) across multiple frontend clients
- Engineering custom threat intelligence dashboards and active transaction monitoring systems

Professional Experience

Founder & Lead Architect (AMTTP)

August 2023 – Present

Derby, United Kingdom

- Engineered and deployed a full-stack decentralised compliance MVP integrating Python microservices, a Next.js administrative dashboard, a Flutter consumer client, and Solidity smart contracts.
- Built a real-time ML inference pipeline capable of processing 1M+ transactions with an ROC-AUC of 0.9879 for anomaly detection, deploying distilled student models for CPU-optimised scaling.
- Published groundbreaking research on Blind-Spot Decomposition Theory (BSDT) through TechRxiv and Zenodo, providing a mathematical geometry for detecting systemic infrastructural collapse.
- Filed a UK patent protecting the core cryptographic and machine learning methodology underlying distributed transaction compliance.

Account Manager (Technical Operations & Vendor Risk)

October 2022 – February 2023

Jumia, Lagos, Nigeria

- Secured multi-million revenue digital payment flows by identifying architectural vulnerabilities in vendor integration protocols and implementing stricter API operational controls for commercial accounts.
- Analysed systemic transaction risks across complex commercial networks, collaborating directly with engineering and finance teams to patch logical flaws that enabled invoice-based and CEO fraud.
- Leveraged data querying and enterprise monitoring tools to isolate anomalous operational behaviour within high-velocity logistics ecosystems, managing strategic vendor relationships to improve platform resilience.

Cyber Security Engineer & Integration Specialist

October 2017 – July 2021

Mctoolu Global Concept, Lagos, Nigeria

- Programmed and executed automated security testing scripts and manual penetration tests (black-box) against critical third-party web application integrations.
- Exploited and subsequently remediated critical OWASP Top 10 vulnerabilities (including complex XSS and SQL injection vectors) in externally exposed client and supplier systems.
- Architected internal security tooling and vendor risk assessment scripts, automating due diligence pipelines for high-risk supply-chain API onboarding.
- Bridged the gap between technical exploits and executive governance, engineering dashboarding systems to visualise security telemetry and enforce NCSC and ISO 27001 control alignment across external infrastructure.

Technical Toolkit

- **Programming & Architecture:** Python (FastAPI), TypeScript (Next.js, Node), Dart (Flutter), Solidity, Shell scripting
- **Data & ML Infrastructure:** XGBoost, Pandas, Memgraph, MongoDB, Redis, Uvicorn
- **Security & CI/CD:** Hardhat, Foundry, Docker, KQL, Zero-Knowledge Proofs (Groth16), API security testing
- **Vulnerability Management Tools:** Nessus, Qualys, Microsoft Defender, bespoke security scripts

Education and Certifications

Bachelor of Technology, Computer Science and Engineering

Ladoke Akintola University of Technology

Selected Qualifications

- CompTIA Security+ (Certified)
- Advanced Software Architecture & Machine Learning (Independent Research & AMTTP Development)

Professional Interests

- Advancing privacy-preserving cryptography (ZK-SNARKs) in enterprise FinTech applications
- Designing low-latency machine learning pipelines for hostile network topologies
- Bridging the technical gap between modern legislative cybersecurity requirements and verifiable software design